

EXAM

This is a 3 hours exam. Class slides and any handwritten material are allowed

I – TRUE, FALSE, UNCERTAIN? (1/3 of the points)

For each of those statements, say whether it is true, false, or uncertain and explain why. You should target around 10 lines per question.

- 1 – When studying the business cycle, the Hodrick-Prescott better fits the data.
- 2 – In a growth model, it is optimal to be along a balanced growth path.
- 3 – (*This question counts the double of any other*) In the Ramsey model, starting from a steady state, an increase in the depreciation rate decreases consumption in the short run and increases it in the long run (use a phase diagram and draw the path of c and k .)
- 4 – Search frictions cause unemployment to be too high.
- 5 – Sustained growth is obtained when production functions have constant returns.
- 6 – In endogenous growth models, better enforcement of intellectual property rights unambiguously raises welfare.
- 7 – Durable goods consumption is a smooth variable in the business cycle.
- 8 – Calibration of an RBC model is not so far from estimation.
- 9 – Consider the following stochastic difference equation: $x_t = \beta E_t x_{t+1} + y_t$, where y_t is an exogenous stationary stochastic process. This equation has a unique stationary solution only if $|\beta| > 1$.

II – GROWTH MODEL WITH HUMAN CAPITAL (1/3 of the points)

Consider an economy with a single infinitely lived household, whose preferences are

$$\int_0^\infty e^{-\rho t} \frac{c_t^{1-\sigma}}{1-\sigma} dt$$

There are two sectors in the economy. Sector 1 produces consumption and capital goods:

$$G(k_{1t}, h_{1t}) = c_t + I_{1t}$$

where $I_{1t} = \dot{k}_t + \delta k_t$ and $k_t = k_{1t} + k_{2t}$. Sector 2 produces human capital:

$$H(k_{2t}, h_{2t}) = I_{2t}$$

where $I_{2t} = \dot{h}_t + \delta h_t$ and $h_t = h_{1t} + h_{2t}$. It is assumed that G and H are constant returns to scale. h_0 and k_0 are given.

There is a government that taxes capital and human capital income. It rebates revenues as a lump-sum transfer T_t . The budget constraint is given by

$$T_t = \tau_k q_{kt} k_t + \tau_h q_{ht} h_t.$$

There is a representative firm in each sector. Firms rent capital and human capital at rental prices q_{kt} and q_{ht} , respectively. Good 1 is the numéraire. The price of good 2 is p_t .

The household's budget constraint is given by:

$$c_t + I_{1t} + p_t I_{2t} = (1 - \tau_k) q_{kt} k_t + (1 - \tau_h) q_{ht} h_t + T_t$$

- 1 – Derive firms and households first order conditions.
- 2 – Derive the household's Euler equation, that can be written as

$$\frac{\dot{c}_t}{c_t} = \frac{r_t - \rho}{\sigma}$$

where r can be expressed in two different ways, depending on whether capital or human capital is considered.

3 – Derive four equations that solve for the balanced growth values of g , z_1 , z_2 , r where g is the balanced growth rate of (c, k, h) and $z_i = k_i/h_i$. Note that p is constant on the balanced growth path. Remember that, with constant returns to scale, marginal products are functions of z_i .

4 – For the special case where $H(k_{2t}, h_{2t}) = Bh_{2t}$, show that taxes on sector 1 do not affect the balanced growth rate. What is the intuition for this result?

III – MORTENSEN-PISSARIDES (1/3 of the points)

Consider the following version of the matching function giving the number of matches M :

$$M = aV^\alpha U^{1-\alpha} \quad (1)$$

where V is the number of vacancies and U is the number of unemployed job-seekers. Assume that $\alpha = 1/2$.

1 – Define $\theta = \frac{V}{U}$ as the measure of labor market tightness. By using the matching function (1), write the rate at which a firm fills a vacant job $\frac{M(V,U)}{V}$ and the rate at which an unemployed worker finds a job $\frac{M(V,U)}{U}$ as a function of θ . Denote the former with $q(\theta)$ and the latter with $p(\theta)$.

2 – Compute $q'(\theta)$ and $p'(\theta)$. Interpret economically the signs of these functions.

3 – Denote the exogenous job destruction rate with δ , wage with w , non-labor market income with b , and the discount rate with r . Write the discounted expected utility of an employed person V_e and the discounted expected utility of an unemployed person V_u where you should use the exit rate from unemployment derived previously.

4 – Denote the output by worker with y and the cost of holding a vacancy with c . Write the discounted expected profits from a filled job Π_e and the discounted expected profits from a vacant job Π_v where you should use the probability of a match occurring derived previously.

5 – Assuming free entry, use the expressions for Π_e and Π_v to derive the firm's labor demand that gives the relationship between θ and the net production $y - w$. This gives the labor demand equation (LD).

6 – Solve the Nash bargaining problem:

$$\max_w \Omega = \frac{1}{2} \log(V_e - V_u) + \frac{1}{2} \log(\Pi_e - \Pi_v)$$

assuming free entry. Write the wage as a function of θ . This gives the wage equation (WE).

7 – Solve for the steady state unemployment where flows in and flows out from unemployment are equal. Use the formulas for $p(\theta)$ and $q(\theta)$ derived previously. This gives the Beveridge curve equation (BC).

8 – By combining the equations (LD) and (WE), one obtains an equation that relates θ to y , b , c , r , δ and a . Derive this equation.

9 – Show how θ , U and w vary with b . Interpret the answers.

10 – Show how θ , U and w vary with a . Interpret the answers.